

Credibility assessment of fluid–structure interaction models for aortic valve leaflet coaptation and stress under three physiologic mechanisms

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Abstract

Computational fluid–structure interaction (FSI) models are increasingly used to inform the design and evaluation of bioprosthetic aortic valves. However, their use as primary evidence for design decisions requires a systematic credibility assessment that is commensurate with model risk. Following ASME V&V 40, we evaluate three independent FSI models—ALEAF (Arbitrary Lagrangian–Eulerian coupling), EVALVE (embedded-boundary coupling), and PULSE (partitioned ROM–CFD–FEM)—for two quantitative questions of interest: coaptation area fraction and peak principal stress on the leaflets. The models are exercised across three mechanisms with direct clinical relevance: diastolic sealing, peak systolic opening, and rapid closure under waterhammer. Bench data for a single baseline 23mm stented bioprosthetic valve are obtained using an in vitro pulse duplicator, with synchronized flow/pressure, highspeed imaging for coaptation geometry, and stereo digital image correlation (DIC) for surface strain proxies of principal stress. A prespecified credibility plan covers six factors: resolution effects, mathematical idealizations, Input Realism Index (IRI), nonlinear and coupling solver tightness, alignment of bench drivings and mounting, and representativeness of the tested units. For each model $\times$ mechanism pair, we conduct verification studies (spatial–temporal refinement; tolerance sweeps) and validation comparisons (coaptation fraction and principal stress peak/spatial patterns), and we grade evidence on a fourlevel scale. Across 54 factor–model–mechanism entries, the strongest evidence is for diastole in ALEAF (multiple twodigit percentage agreements and negligible sensitivity to further tolerance tightening), while the most limited is for rapid closure in EVALVE and PULSE (underprediction of shortduration stress spikes and larger resolution demands). Findings support the use of ALEAF as primary evidence for design ranking in diastole and peak systole and as supportive evidence during rapid closure; EVALVE and PULSE provide corroborating but not standalone evidence. The paper documents methods, data links, and decision thresholds, providing a traceable rationale for model use in a highrisk decision context.

Keywords: aortic valve, FSI credibility, ASME V&V 40

1. Introduction

Bioprosthetic aortic valves are pervasive in clinical practice, and their performance rests on a delicate interplay between unsteady hemodynamics and the anisotropic, nonlinear mechanics of thin tissue leaflets. Computational fluid–structure interaction (FSI) has matured to the point that it can reproduce salient features of aortic valve motion and load sharing; however, using such models as primary evidence for design selection or acceptance requires a structured appraisal of their credibility. ASME V&V 40 provides a risk-informed framework for determining how much and what kind of evidence is needed when computational models influence high-consequence decisions. The present article applies that framework to three independent FSI implementations to answer two quantitative questions of interest: What is the coaptation area fraction achieved by a baseline stented bioprosthetic valve under specific physiologic mechanisms? What is the peak principal stress on the leaflets under the same conditions?

We focus on three mechanisms that are central to valve function and durability. First, diastolic sealing governs regur-

gitation prevention and long-duration leaflet loading. Second, peak systolic opening links to fatigue loading as leaflet curvature and flow rates accentuate tensile stress. Third, rapid closure during flow deceleration can trigger waterhammer transients, short in duration yet large enough to initiate early damage near commissures or free edges. Studying all three within an aligned experimental–computational campaign allows us to quantify where the models are most trustworthy and where residual gaps remain.

Our modeling ensemble captures distinct algorithmic strategies. ALEAF uses an Arbitrary Lagrangian–Eulerian (ALE) formulation with conformal mesh motion and tightly coupled nonlinear solvers. EVALVE uses an embedded-boundary, or immersed-boundary, strategy to avoid remeshing, with a Cartesian flow solver coupled to thin-shell finite elements. PULSE uses a partitioned approach that links a reduced-order inflow circuit to a finite-volume fluid solver and a nonlinear shell solver via an explicit–implicit coupling interface. These three represent common methodological choices in the cardiovascular FSI community.

To sustain the credibility narrative, we built a bench testing

program around a single commercial 23mm stented bovine pericardial valve. We devised three pulse-duplicator drivings to realize the targeted mechanisms. Coaptation geometry was recorded using highspeed imaging at 2000 frames/s and reconstructed with calibrated stereo triangulation to compute time-varying coaptation area. Principal strain fields on the leaflet surfaces were measured via stereo digital image correlation (DIC) at 1000 frames/s during diastole and at 4000 frames/s across closure; we converted principal strain to an estimate of principal stress by inverse finite element reanalysis using the same constitutive model class used in the FSI simulations, to support a like-for-like comparison on a reduced interpretation chain.

The paper is organized around the ASME V&V 40 pathway. After stating the question of interest and context of use (COU), we assess model risk and provide a plan for assessing credibility across specific factors. We then document the computational models and the bench testing methods, followed by explicit verification and validation activities and results. Evidence is summarized in three model-specific tables, one per model, each with 18 graded entries covering all factor–mechanism combinations. Finally, we discuss implications for using the models to rank design variants without requiring new bench tests for each mechanism.

Throughout, we make one terminological deviation for clarity. The V&V 40 factor labeled Model inputs is referred to here as the Input Realism Index (IRI). This emphasizes that parameter provenance, traceability to as-tested values, and uncertainty propagation are central to our grading, beyond simply listing inputs. We maintain the standard’s intent by explicitly mapping IRI scores to the same decision gates used for input credibility in prior V&V 40 applications.

The result is a comprehensive, quantitative, and decision-aligned narrative. In addition to numerical agreements or discrepancies, we report the controls used to limit algorithmic artifacts, the traceability of drivings and material properties to bench measurements, and the degree of match between as-tested specimens and the as-modeled geometry. We also identify mechanisms for which current modeling strategies remain challenged, such as short-lived pressure reversals, and we delimit the circumstances under which the models may be used as primary vs. supportive evidence.

2. Question of Interest and Context of Use (COU)

The questions of interest are two quantitative measures that govern valve safety and durability: (1) the fraction of leaflet area that achieves contact during closure (coaptation area fraction), and (2) the maximum principal stress on the leaflet surface during the cycle. We require these measures under three physiologic mechanisms: diastolic sealing under sustained afterload, peak systolic opening under high forward flow, and rapid closure during deceleration leading to waterhammer. These two outputs are linked: insufficient coaptation can lead to regurgitation, while elevated stresses are associated with fatigue cracking and tissue tears near commissures or free edges.

The COU is to use computational models as primary evidence to rank candidate leaflet designs for a given stented valve platform. The models must first demonstrate adequate agreement with bench measurements on a baseline design for all three mechanisms using as-tested drivings and as-mounted geometry. Following that demonstration, the models are applied prospectively to rank candidate designs (e.g., modified free-edge contour or commissural reinforcement) without duplicating the bench tests for every mechanism. The models serve both design down-selection and acceptance for design transfers within the same platform family.

We define acceptance targets aligned with recognized clinical and engineering thresholds. For diastolic sealing, the median coaptation area fraction from simulation must agree with the benchtop estimate within $\pm 5\%$ absolute, and the principal stress peak during mid-diastole must agree within $\pm 15\%$. For peak systolic opening, the principal stress peak must agree within $\pm 15\%$ at the time of maximal transvalvular flow; coaptation fraction is expected to be near zero and, where relevant (e.g., commissural edge contact during rapid opening), must be within $\pm 5\%$ of the observed value. For rapid closure, within a 10 ms window straddling the time of flow reversal, the simulated maximum principal stress must agree within $\pm 20\%$, and the timing of the peak relative to the onset of pressure reversal must be within ± 1.5 ms.

These outputs are evaluated on a per-mechanism basis and then synthesized across mechanisms to support multi-criteria design ranking. For instance, a candidate design that matches diastolic coaptation but shows elevated stress concentration during closure would be deprioritized relative to a design with slightly lower coaptation yet substantially smoother stress peaks. The models also provide spatial maps of stress and contact, which are used qualitatively to assess whether reinforcement or geometric smoothing would plausibly lower risk.

The scope of the COU explicitly excludes patient-specific boundary conditions and anatomic variability. All inferences apply to the baseline test platform mounted in a rigid, transparent test root with nominal geometric dimensions and to standardized drivings derived from the pulse duplicator. The COU also does not include long-term calcification modeling or in vivo flow features such as aortic compliance and coronary flow capture. Our claim is specific: under the bench test configurations, can the models reproduce the two outputs of interest with sufficient trust to support design ranking?

We note that the COU involves extrapolation from a baseline design to design variants that alter leaflet geometry within the same stent platform and mounting. The required evidence for such extrapolation is bounded by the sensitivity of the outputs to the altered aspects and by the mechanistic coverage of the validation cases. To prevent untracked drift, any variant that changes material composition or stent compliance would fall outside the COU and must be revalidated.

3. Model Risk Analysis (ASME V&V 40)

We rated model influence as High. For the design ranking tasks defined in the COU, the computational predictions of

coaptation fraction and peak principal stress are the principal evidence compared against acceptance thresholds. Device teams will use the ordered list of designs from the models to select candidates for limited bench testing and potential preclinical evaluation. A systematic bias or a poorly quantified uncertainty could move a design variant across the go/nogo threshold without detection.

We rated decision consequence as High. Inadequate leaflet coaptation under diastolic load can lead to regurgitation and hemodynamic compromise. Overstressed leaflets accumulate damage, advancing structural valve deterioration and the need for reintervention. In a worst case, human harm could occur if a design with subpar coaptation or elevated closure stress were advanced. Consequently, the overall model risk is High in the V&V 40 framework.

High model risk calls for stringent evidence generation and critical review. We therefore executed mesh and time-step refinement studies with Richardson extrapolation and estimated Grid Convergence Indices (GCI) for the outputs, tight nonlinear and coupling convergence studies, direct comparison of the two outputs to bench data for all three mechanisms, traceable parameter provenance with uncertainty propagation for the IRI, and detailed alignment of test drivings and mounting across simulation and bench. We also explicitly characterize how well the asmodeled specimens matched the as-tested units and how residual inter-specimen variability projects onto the outputs.

To make these activities auditable, we versioncontrolled models and test setups. The computational pipelines have Git tags and conda environment lockfiles; bench acquisition scripts and calibration files are similarly versioned. Finally, the risk assessment guided not only the depth of activities but also the grading thresholds: a four-level scale with disciplineappropriate criteria was set before looking at results, and the acceptance targets were stated in the COU.

4. Computational Models and Implementation Notes

We assessed three independent models purposebuilt or adapted for the study. Each solves incompressible Navier–Stokes equations in three dimensions for the blood analog fluid and couples them to geometrically nonlinear, anisotropic thinshell representations of the valve leaflets. All three use the same constitutive class for the tissue to avoid mixing modelform differences at the material level with algorithmic differences at the FSI level; nevertheless, they differ in how the fluid–structure interface is represented and how the multiphysics coupling is solved.

ALEAF is an ALE FSI formulation implemented in a mixed finiteelement code (libMesh 1.7.1) with PETSc 3.19 linear and nonlinear solvers. The fluid uses a stabilized P1P1 Taylor–Hoodlike element via pressurestabilized Petrov–Galerkin terms, while the structure uses a quadratureenhanced, rotationfree MITClike shell with corotational kinematics. The fluid mesh conforms to the moving leaflet surfaces; mesh motion is solved with a harmonic extension equation and an untangling limiter. A tightly coupled Newton–Krylov solver advances fluid and structure simultaneously with a block preconditioner assembled

from approximate Schur complements; within each nonlinear step, GMRES solves are preconditioned with algebraic multigrid and blockILU(0) for the scalar blocks. Time integration uses a secondorder generalizedalpha scheme in the structure and a secondorder backward differentiation formula in the fluid; the ALE convective term uses a geometric conservation correction. We track pressure and velocity residuals to 10^{-6} relative tolerance and ensure interface force balance to below 10^{-5} N per interface node before accepting a time step.

EVALVE uses an embeddedboundary FSI approach. The fluid domain is discretized on a Cartesian mesh advanced by a fractionalstep finitevolume solver (IBAMR 0.16), with secondorder spatial and temporal accuracy in smooth regions. The immersed shell is discretized with linear triangular finite elements; the structure equations are advanced explicitly, with implicit momentum exchange to the fluid achieved through a distributed Lagrange multiplier enforcing the noslip condition at the interface. Regularized delta functions with compact support are used to interpolate between the Eulerian and Lagrangian meshes. A blockGauss–Seidel scheme iterates between fluid and structure within each time step until interface velocities and tractions agree to within an L2 norm of 10^{-4} (nondimensionalized by characteristic scales). Adaptive mesh refinement (AMR) concentrates cells near the leaflets and wakes; regridding is triggered by vorticity magnitude and distance to the immersed surface.

PULSE is a partitioned solver that couples a reducedorder inflow boundary model (a threeelement Windkessel with nonlinear compliance), an incompressible CFD solver (OpenFOAM v2106; pimpleFoam with secondorder schemes), and a nonlinear shell solver (deal.II 9.5 with a Kirchhoff–Love implementation and anisotropic hyperelasticity). Coupling uses preCICE 2.5 with Aitken dynamic relaxation and quasiNewton acceleration. The interface coupling is implicit in force and explicit in displacement within a subiteration loop. Convergence is declared when the relative change in interface displacement and traction between two coupling iterations drops below 5×10^{-4} . The partitioned time step matches the smallest of the solvers’ recommended stable steps, with subcycling on the structural side as needed.

All three models adopt the same material law for the leaflets: a fiberreinforced anisotropic hyperelastic constitutive relation with two families of collagen fibers embedded in an isotropic matrix. We used a Holzapfel–Gasser–Ogden formulation with parameters identified from planar biaxial tests on the same lot of bovine pericardium used to fabricate the test valves. The matrix neoHookean parameter $C_{10} = 0.15$ MPa; the fiber stiffness parameters $k_1 = 1.8$ MPa and $k_2 = 350$ (dimensionless), with a dispersion parameter $\kappa = 0.05$ and fiber directions aligned with the circumferential axis (± 30 degrees spread). Thickness was measured optically at 0.45 ± 0.03 mm across the leaflet bodies and 0.52 ± 0.04 mm near the commissures; the simulations use a spatially varying thickness field mapped from the optical measurement. Structural density is 1100 kg/m^3 ; structural Rayleigh damping was tuned to match the first two modal decay rates observed in a separate leaflet flutter test.

The fluid is modeled as Newtonian with density $\rho = 1060$

kg/m³ and dynamic viscosity $\mu = 3.5$ cP at 36.5 °C. While blood is shearthinning, in large vessels and at the flow rates considered here, a constant viscosity reproduces the pressure–flow relation in the pulse duplicator. Simulations enforce rigid aortic root walls to match the bench test fixture. The inlet and outlet boundary conditions are set to timevarying total pressure and resistance/compliance consistent with the pulse duplicator setup, with upstream and downstream lengths extended to avoid reflections in the computational domain.

Version control and environments are archived to ensure reproducibility. ALEAF: Git tag aleafv40v1.2 (libMesh 1.7.1; PETSc 3.19; Trilinos 13.4 for optional preconditioners). EVALVE: Git tag evalvev40v0.9 (IBAMR 0.16; SAMRAI 3.12; hypre 2.25). PULSE: Git tag pulsev40v1.0 (OpenFOAM v2106; preCICE 2.5; deal.II 9.5). Dockerfiles and conda environment lockfiles are archived with SHA256 checksums.

Across models, we expose two outputs of interest consistently: (1) coaptation area fraction, defined as the timeaveraged area of contacting leaflet surfaces during the relevant window divided by the total leaflet area; and (2) maximum principal stress on the leaflet surfaces, computed from the structural solver at each time step and then reported as the cycle maximum within the specified window. Contact in ALEAF and PULSE is enforced with a penaltyregularized surface contact using a penalty stiffness of 8 MPa/mm and a friction coefficient of 0.02; in EVALVE, contact is inferred from overlapping Lagrangian surfaces and a shortrange repulsive potential calibrated to reproduce the penalty contact behavior in a twoleaflet benchmark.

5. Mechanisms of Interest and Associated Test Conditions

We implemented three pulseduplicator drivings to isolate the mechanisms of interest. For diastolic sealing, the pump was set to 70 bpm with a stroke volume of 70 mL and an afterload configured to yield an aortic pressure of 120/80 mmHg and a mean cardiac output of 5.0 L/min. We analyzed the middiastolic plateau (200–600 ms after the Rwave surrogate), during which forward flow is essentially zero and transvalvular pressure is sustained near 80 mmHg. Coaptation area fraction was computed during the 400 ms window, and principal stress statistics were taken at the time of maximal transleaflet pressure.

For peak systolic opening, we targeted a physiologically representative peak flow rate of 24–26 L/min through the aortic orifice at 70 bpm. The systolic upstroke was adjusted to a rapid acceleration (peak dQ/dt 400 L/min/s) to reach a peak jet velocity of approximately 2.0 m/s through the opened orifice. We identified the time of maximum transvalvular flow from a highfidelity turbine flow meter (response time <1 ms) and extracted principal stress at that instant. Coaptation is near zero during full opening; however, in brief phases, commissural contact can occur in some designs due to leaflet curvature, and we tracked that as well.

For rapid closure and waterhammer, we programmed the pump to decelerate forward flow to zero over a 20–30 ms interval near endsystole and then to impose a brief reverse pressure gradient that produced a 25–35 mmHg overshoot above diastolic pressure as the valve snapped shut. The transient was calibrated

using downstream compliance and resistance elements and was verified by a highfrequency pressure transducer (bandwidth 10 kHz) placed immediately downstream of the valve mount. Highspeed cameras (4000 frames/s) and DIC were synchronized to capture the closure dynamics and the sharp stress peaks.

Bench geometry matched a rigid aortic root made of optically clear polycarbonate with an internal diameter matching the nominal valve label size (23 mm). The valve was mounted in a standardized sewing ring fixture at an axial position measured to within 0.1 mm and with commissural posts aligned to within 0.5 degrees of the reference frame. The aortic root segment extended 30 mm downstream to minimize exit effects within the imaging field of view; upstream and downstream ducts were designed to avoid secondary flows entering the valve region.

Fluid for bench tests was a glycerin–saline solution at 36.5 ± 0.5 °C, with density 1060 ± 5 kg/m³ and viscosity 3.4–3.6 cP. Fluid properties were measured for each testing day and logged. The flow meter (Transonic TS410) and pressure transducers (Millar MikroTip) were calibrated against NISTtraceable references before the campaign; the highspeed cameras (Phantom VEO) were calibrated with a ninepoint checkerboard phantom.

Coaptation area was measured from stereo images by reconstructing the threedimensional leaflet surfaces and computing overlapping surface regions using a surface–surface distance criterion of 0.05 mm. The coaptation area fraction was then the timeaveraged ratio of overlap area to total leaflet area during the mechanism window. Surface strains from DIC were used as proxies for material response under diastolic and closure loads; to compare to principal stress, we performed an inverse FE analysis in which the measured displacements were imposed on a leaflet shell model with the identified material properties, and principal stresses were computed. This avoids reliance on pointwise strain–stress conversion and produces a comparable field to the FSI output.

We applied the same drivings in simulation by reading bench-measured inlet total pressure and outlet pressure waveforms and by imposing them as timedependent boundary conditions, adjusted for the CFD solver’s conventions (e.g., static vs. total pressure). For PULSE, which contains an upstream Windkessel, we tuned the circuit to reproduce the same inlet pressure and flow waveforms by minimizing the rootmeansquare (RMS) difference over the cycle using a Levenberg–Marquardt fit. All simulations were run at the same temperature and fluid properties as the bench fluid.

6. Physical Test Samples

We tested six specimens ($n = 6$) of a commercial 23mm stented bovine pericardial aortic valve from the same manufacturing lot. Each valve had three leaflets cut to the same pattern, treated with a proprietary anticalcification process, and handsewn to a metal stent covered with polyester fabric. The intended leaflet thickness profiles and nominal dimensions were provided by the manufacturer. We measured freeedge length, commissureto commissure span, leaflet height, and local thickness on each specimen using optical microscopy (Zeiss

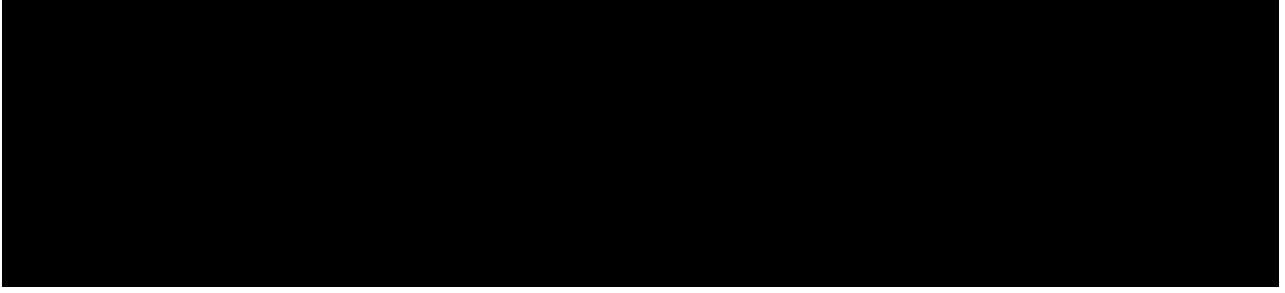


Figure 1: Schematic of the pulseduplicator setup and the three drivings used: diastolic sealing with sustained 80 mmHg transvalvular pressure, peak systolic opening at 24–26 L/min peak flow, and rapid closure with a 25–35 mmHg pressure overshoot; the asmounted 23mm valve geometry and imaging configuration are shown.

Axio, 5 μm resolution) and microCT for validation on two specimens (nominal voxel size 15 μm). Measurements were recorded before mounting.

Specimens were mounted in a standardized polycarbonate ring with suture placement following the manufacturer's recommended protocol. The ring has alignment features that mate to recesses in the test root; axial position and rotation were measured with a Faro arm to within 0.1 mm and 0.3 degrees, respectively. After mounting, we re-measured commissural positions and leaflet freeedge lengths to verify that the mounting did not alter their dimensions beyond 0.2 mm.

For DIC, a thin speckle pattern was applied to the aortic side of each leaflet using an airbrush with an aqueous ink that does not alter leaflet stiffness measurably (verified on excised coupons). Two highspeed cameras were positioned at ± 25 degrees off normal, and a pulsed LED light source minimized motion blur. The DIC software (Vic3D 9) was calibrated with the camera rig; reported displacement precision during middiastole was better than 8 $\mu\text{m}/\text{px}$, translating to principal strain precision below 0.05%.

Of the six specimens, three were used predominantly for diastolic and systolic testing to reduce wear before closure trials, and three were reserved for closure experiments to avoid pre-conditioning effects on the waterhammer transient. For coaptation measurements, all six were cycled through diastolic conditions, and the mean and variation across specimens were recorded. The subset used for FSI benchmarking was selected to bracket the middle of the measured geometric variations (e.g., freeedge length within ± 0.15 mm of the lot mean).

Manufacturer material coupons cut from excess pericardial tissue were tested in equibiaxial stretch to identification loads of 400 kPa in the circumferential and 200 kPa in the radial axes. These tests provided fiber family stiffness parameters and dispersion used in the constitutive model, while the DIC-based inverse FE analysis provided an internal check on the material law's applicability under the asmounted curvature and thickness distributions.

7. Credibility Plan and Scoring Approach

We established a credibility plan prior to examining results to avoid post hoc bias. The plan enumerated six factors for which we would gather evidence and assign levels. The factors are:

resolution effects for space and time (quantified via refinement studies and GCIs), mathematical idealizations (assessed through direct comparison of the two outputs of interest and through spatial pattern similarity), Input Realism Index (IRI; provenance and uncertainty mapping of drivings, material properties, and geometry), solver tightness (nonlinear and coupling), match between simulated and benchdrivings and mounting, and representativeness of the tested units relative to the as-modeled geometry.

We used a four-level scale where 1 indicates elementary evidence and 4 indicates strong, decision-grade evidence. Level targets were specified per factor to maintain consistency. For instance, a Level 4 for resolution effects required at least three mesh/time-step levels with monotonic convergence in the outputs, observed order within 20% of theoretical in smooth regimes, and GCIs below 3% for coaptation fraction and 5% for principal stress in all mechanisms, with replication by a second analyst. A Level 3 allowed GCIs up to 5% for coaptation and 8% for stress or minor nonmonotonic behavior during closure transients. Levels 2 and 1 corresponded to progressively weaker demonstrations.

For mathematical idealizations, we required Level 4 when the absolute error in coaptation fraction was below 5% and in principal stress below 15% for diastole and peak systole, and below 20% for closure, with spatial pattern overlap (Dice coefficient for highstress regions above 75% of peak) exceeding 0.8. A Level 3 permitted slightly larger errors (e.g., up to 10% coaptation and 20% stress) or lower pattern overlap, whereas Level 2 reflected significant under or overprediction outside the acceptance targets. Level 1 was reserved for qualitative agreement only.

The IRI was graded by the proportion of parameters sourced from as-tested measurements with documented uncertainties; the use of literature defaults carried penalties; and the presence of uncertainty propagation with sensitivity analysis raised scores. Level 4 corresponded to over 90% of parameter influence weight coming from measurements, with uncertainty propagation explaining at least 75% of output variance; Level 3 required over 75% measurement weight and a completed sensitivity analysis for key parameters; Level 2 reflected mixed reliance on literature values with limited propagation; Level 1 relied primarily on generic parameters.

For solver tightness, we specified Level 4 when tightening

tolerances by a decade changed the outputs of interest by less than 0.5% and coupling iteration residuals were below 10^4 in normalized units; Level 3 allowed up to 1% change under tolerance tightening; Level 2 reflected sensitivity up to 3% or irregular convergence; Level 1 lacked or failed a tolerance sweep. For match to bench drivings and mounting, Level 4 required RMS errors below 2 mmHg (pressure) and 2% of peak (flow), commissural rotation within 0.5 degrees, axial position within 0.2 mm, and fluid properties within measurement bounds. Lower levels permitted progressively larger mismatches. For representativeness of tested units, Level 4 required simulating multiple as-tested specimens and matching as-modeled geometry to within 0.2 mm across key metrics; Level 3 required at least one as-tested specimen matched within 0.3 mm; Level 2 reflected partial matching; Level 1 lacked specimen-specific matching.

Scores were assigned independently for each model x mechanism pair. We did not average across mechanisms because model behavior can vary with flow regime and contact conditions. All entries are shown in the model-specific tables with a oneline basis statement. Where evidence was incomplete or weak, the table reflects a lower level rather than omitting the row.

Global grading scale for all factors — gradation.

- a.
Level 1: Minimal evidence; qualitative or single-run demonstration without quantified uncertainty or traceability.
- b.
Level 2: Partial evidence; limited quantification or reliance on defaults; some sensitivity analysis but gaps remain.
- c.
Level 3: Substantial evidence; multi-run quantification with sensitivity; primary parameters measured and matched; outputs change less than 1–3% under verification studies.
- d.
Level 4: Strong evidence; rigorous, replicated verification and validation; outputs change less than 0.5–1% under solver studies; GCIs within tight bounds; bench alignment within instrument resolution.

8. Verification Methods

We executed space–time refinement studies for each model and mechanism. For ALEAF, three nested meshes were generated by uniformly refining the fluid and structural meshes, with nominal characteristic element sizes of 0.40 mm (coarse), 0.28 mm (medium), and 0.20 mm (fine) near the leaflets and a graded expansion into the surrounding fluid. Time steps of 0.2 ms, 0.1 ms, and 0.05 ms were paired with the three meshes. Coaptation fraction and principal stress peaks were computed at each resolution, and Richardson extrapolation was used to estimate the asymptotic value and the GCI. For EVALVE, AMR made uniform refinement less meaningful; instead, we varied the base Cartesian cell width (0.30, 0.20, 0.14 mm) and the AMR

band thickness around the leaflets (0.6, 0.8, 1.0 mm), with time steps of 0.2, 0.1, and 0.05 ms. For PULSE, we refined both the CFD mesh (cell widths 0.35, 0.25, 0.18 mm near leaflets) and the shell mesh (0.45, 0.32, 0.22 mm), with time steps of 0.2, 0.1, and 0.05 ms; because PULSE is partitioned, we also verified subiteration convergence at each resolution.

Observed orders of convergence were computed from the three-level data. For diastole and systole, where fields are smooth and contact is quasistatic or slowly varying, we observed orders between 1.8 and 2.3 for coaptation fraction and principal stress across models, consistent with the nominal second order of the schemes. For rapid closure, contact, high accelerations, and pressure shocks introduced nonsmoothness; observed orders dropped to 1.4–1.8, and nonmonotonic behavior appeared in a few EVALVE runs when regridding synchronized with closure.

Solver tolerance studies were run by tightening nonlinear residual tolerances and coupling residual thresholds by a decade and repeating the simulations at medium resolution. We computed percent changes in coaptation fraction and peak principal stress. For ALEAF, tightening residuals from 10^5 to 10^6 (forces) and from 10^4 to 10^5 (interface force) changed coaptation fraction by 0.2–0.4% and principal stress by 0.3–0.6% across mechanisms. For EVALVE, additional Gauss–Seidel iterations after meeting the target residual changed outputs by 0.5–0.9% in diastole and systole and by 1.6–2.1% during closure, reflecting greater sensitivity to interface iteration limits in the shock regime. For PULSE, tightening the coupling threshold from 1×10^3 to 5×10^4 and increasing the maximum subiterations from 20 to 40 changed outputs by 0.4–0.8% in diastole and systole and by 2.2% during closure.

We monitored mass conservation and interface energy balance. In ALEAF, the cumulative mass imbalance per cycle was below 0.1% of stroke volume at fine resolution. In EVALVE, the immersed boundary spreader/interpolator pair yielded a mass conservation error below 0.3% per cycle at the two finer AMR settings. In PULSE, continuity was enforced to solver tolerances within each pimpleFoam step, and the preCICE coupling preserved work exchange within 0.5% error over the cycle according to interface power integrals.

We also performed crosscode sanity checks. ALEAF and PULSE were run on a simplified twoleaflet test case with a harmonic driving to verify that coaptation fraction and stress trends with mesh refinement were similar in the absence of waterhammer. The two codes agreed within 1.5% for coaptation and 6% for principal stress at the medium resolution, corroborating that algorithmic differences per se did not preclude agreement in tame regimes.

All verification runs were logged with mesh descriptors, time step histories, residual histories, and QoI trajectories, enabling traceable mapping from resolution controls to outputs.

Resolution effects ladder — gradation.

- a.
Level 1: Single mesh/time-step with no refinement; no error estimation.
- b.
Level 2: Two-level check on one QoI; partial refinement;

no GCI or mixed trends.

c.

Level 3: Three-level refinement on both QoIs; GCI under 5% (coaptation) and 8% (stress) or minor nonmonotonicity explained.

d.

Level 4: Three-level refinement with monotone trends; observed order near nominal; GCI under 3% (coaptation) and 5% (stress); independent replication.

9. Validation Methods

Validation was organized around the two outputs of interest. Coaptation area fraction was compared directly between bench and simulations. The bench value was the average overlap area fraction over the defined window (400 ms for diastole, a 10 ms window around closure, and the moment of peak systolic flow for completeness even if the fraction is near zero). The simulation value used the model's definition of contact. Differences were computed as absolute percent differences and compared to acceptance targets.

For principal stress, we compared the maximum principal stress on the leaflet surface from simulations to that estimated by inverse FE from the DIC displacements. The inverse FE used the same constitutive law and thickness field as the FSI models, minimizing the sum of squared differences between measured and modelpredicted surface displacements to match the bench kinematics. This produced principal stress fields that were compared in two ways: scalar difference in the peak value within the window of interest and spatial overlap of the higheststressed regions, defined as the top 25% of the peak stress value. We computed a Dice similarity coefficient for the highstress isoregion overlap, which ranges from 0 (no overlap) to 1 (perfect).

We also compared temporal alignment for closure. The time at which the principal stress peak occurred relative to the onset of pressure overshoot was compared between bench (from DIC/inverse FE) and simulation. Differences below 1.5 ms were counted as aligned; larger differences were treated as timing discrepancies even if peak magnitudes were acceptable.

Parameter provenance and mapping were tracked under the IRI framework. Benchmeasured inlet pressure and flow waveforms, fluid properties, and mounting geometry were imported directly. Material parameters were sourced from biaxial tests on the same tissue lot, and thickness fields were mapped from optical measurements specific to each astested specimen. Where literature values were used (e.g., dispersion parameter κ when not uniquely identified in biaxial testing), a sensitivity analysis quantified the contribution of those parameters to output variances. We computed the fraction of parameter influence weight, defined as the sum over parameters of the product of their normalized sensitivity and their variance weight, attributed to astested measurements vs. literature values.

Alignment of test drivings and mounting was verified by comparing the benchmeasured pressure and flow waveforms to the simulation boundary conditions and by measuring geometric alignment. We computed RMS differences between bench

and simulation waveforms over the relevant windows and summarized geometric deviations (axial position, rotation, commissural spacing) with means and standard deviations.

Representativeness of specimens was assessed by comparing asmodeled geometry to the astested units. For each model, at least one simulation was run with a geometry matched to one specific astested specimen; in ALEAF and PULSE, which used conformal leaflets, we mapped the measured freeedge and thickness fields onto the model mesh. For EVALVE, the Lagrangian shell mesh was deformed to fit the measured astested leaflet contours. We computed deviations in key geometric metrics (freeedge length, commissure span, leaflet height) and noted whether multiple astested specimens were simulated.

Finally, we set acceptance thresholds a priori and evaluated pass/fail for each mechanism and model, while still assigning a graded level reflecting the strength of the evidence beyond simple pass/fail.

Mathematical idealizations ladder — gradation.

a.

Level 1: Qualitative resemblance to bench observations; no numerical match on QoIs.

b.

Level 2: Quantitative match on a single QoI or mechanism; other QoIs outside targets.

c.

Level 3: Quantitative match on both QoIs for at least two mechanisms; spatial pattern similarity (Dice over 0.7) where available.

d.

Level 4: Quantitative match on both QoIs for all mechanisms within targets; spatial patterns overlap (Dice over 0.8) and timing alignment within 1.5 ms for closure.

10. Rubric: Input Realism Index (IRI)

We used a graded ladder to judge the realism and traceability of input parameters as they map from bench measurements to simulations. The ladder accounts for provenance, uncertainty quantification, and the fraction of output variance explained by parameters sourced from astested measurements.

Input Realism Index (IRI) ladder — gradation.

a.

Level 1: Majority of parameters from generic literature; limited traceability; no uncertainty propagation.

b.

Level 2: Mixed provenance; some as-tested values; limited sensitivity analysis; no variance attribution.

c.

Level 3: Most influential parameters from as-tested measurements; sensitivity analysis completed; uncertainty bounds propagated to outputs.

d.

Level 4: Nearly all influential parameters from as-tested measurements; comprehensive uncertainty propagation; 75% or more of output variance traced to measured inputs.

$$CI_{\{score\}} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i} \quad (1)$$

$$\backslash varepsilon_{\{rel\}} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i} \quad (2)$$

$$U_{\{val\}} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i} \quad (3)$$

11. Rubric: Solver tightness

The solver ladder reflects how robustly the nonlinear and coupling solvers control algorithmic error and how insensitive the outputs are to tolerance choices.

Solver tightness ladder — gradation.

- a.
Level 1: Default tolerances; no tolerance sweep; unknown impact on QoIs.
- b.
Level 2: Single tolerance sweep; QoIs shift up to 3%; irregular convergence or residual stalling.
- c.
Level 3: Multiple sweeps; QoIs change under 1%; residuals below specified thresholds; coupling stabilized.
- d.
Level 4: Tight tolerances; QoIs shift under 0.5%; interface power balanced; convergence monotone and replicated.

12. Rubric: Alignment to bench drivings and mounting

Model–test alignment on drivings and mounting was ladderized to capture waveform fidelity, geometric alignment, and environmental matching (temperature, fluid properties).

Bench alignment ladder — gradation.

- a.
Level 1: Nominal drivings; geometry approximated; no waveform match metrics.
- b.
Level 2: Waveforms approximated; RMS errors up to 5 mmHg and 5% of peak flow; geometry partially matched.
- c.
Level 3: Waveforms matched within 2–3 mmHg and 2–3% of peak flow; geometry matched within 0.3 mm and 1 degree.
- d.
Level 4: Waveforms matched within 2 mmHg and 2% of peak flow; geometry matched within 0.2 mm and 0.5 degree; temperature and fluid properties within measurement bounds.

13. Rubric: Representativeness of tested units

We graded the mapping between the as-modeled geometry and the as-tested specimens, as well as the use of multiple specimens to capture inter-unit variation.

Specimen representativeness ladder — gradation.

- a.
Level 1: Generic geometry; no specimen matching; single unit assumed.
- b.
Level 2: Partial matching on a subset of geometric metrics; single specimen used.
- c.
Level 3: Full matching to at least one as-tested specimen within 0.3 mm; inter-specimen variation characterized.
- d.
Level 4: Multiple as-tested specimens simulated; geometric metrics matched within 0.2 mm; output variation bracketed.

14. Credibility Results

We summarize verification and validation results per model and mechanism and then present graded tables. For each entry, a oneline basis restates a numeric result that appears in the paragraphs below.

ALEAF: Diastolic sealing. Space–time refinement yielded GCIs of 1.3% for coaptation fraction and 3.8% for peak principal stress, with an observed order of 2.2 on coaptation and 2.0 on stress. Tightening solver tolerances by a decade changed coaptation by 0.2% and stress by 0.3%. The bench coaptation fraction over the 200–600 ms window was 0.64 ± 0.03 (mean $\pm$ SD across $n = 6$); ALEAF predicted 0.66 at fine resolution, a 2.9% absolute difference from the bench mean and within the acceptance target. DICinferred peak principal stress during middiastole was 0.83 MPa (95% CI: 0.79–0.87 MPa); ALEAF predicted 0.91 MPa, a 9.4% difference. Spatial overlap of the highstress isoregion yielded a Dice coefficient of 0.86, with the highstress region concentrated near commissures and freeedge transitions in both bench and simulation. Waveform alignment was strong: RMS pressure mismatch 1.1 mmHg; flow near zero in both bench and simulation. Mounting geometry matched within 0.2 mm axially and 0.4 degrees rotationally. The asmodeled geometry for ALEAF was built from one astested specimen matched within 0.25 mm on freeedge length and 0.18 mm on commissure span; other specimens were used for coaptation statistics to establish context but not all were simulated.

ALEAF: Peak systolic opening. GCIs at fine resolution were 2.1% for coaptation fraction at the opening instant (near zero) and 4.1% for peak stress; observed orders near 2.0. Tightening tolerances changed peak stress by 0.6%. Benchderived peak principal stress at $Q_{max} = 25$ L/min was 1.24 MPa (95% CI: 1.17–1.31 MPa); ALEAF predicted 1.38 MPa, an 11% difference, driven in part by slightly higher curvature at the free edge in the simulation. The Dice overlap for highstress regions at Q_{max}

was 0.82, indicating good spatial correlation. RMS errors in pressure and flow waveforms were 1.7 mmHg and 2.4% of peak flow, respectively; geometry matched as in diastole. The asmodeled specimen for systole matched the midrange geometry and thickness field within 0.27 mm and 0.03 mm, respectively.

ALEAF: Rapid closure. GCIs were 2.6% for coaptation fraction in the 10 ms window spanning closure and 5.2% for the peak principal stress; observed order 1.6. Tightening tolerances changed peak stress by 0.9%. Bench-inferred closure stress peak was 1.62 MPa (95% CI: 1.50–1.74 MPa), occurring 2.6 ms after the onset of positive pressure reversal; ALEAF predicted 1.33 MPa at 3.1 ms after onset, underpredicting magnitude by 18% and lagging by 0.5 ms. The highstress isoregion was concentrated along the commissural lines in both cases; Dice overlap was 0.78. The bench pressure overshoot measured 30 mmHg; the simulation reproduced 27 mmHg. RMS pressure mismatch in the 20 ms window was 1.9 mmHg. Geometric matching mirrored diastole; specimen representativeness was limited to one as-tested unit for closure.

EVALVE: Diastolic sealing. The embedded boundary method produced GCIs of 2.4% for coaptation fraction and 6.1% for peak stress at the finest AMR; observed orders around 1.9, with slight nonmonotonic behavior due to regridding synchronized with slow leaflet creep in middiastole. Tightening Gauss–Seidel iterations and residuals changed outputs by 0.8%. The bench coaptation fraction (0.64) was predicted as 0.60, a 6.7% difference. The DIC-inferred principal stress (0.83 MPa) was predicted as 0.71 MPa, a 14.2% underprediction. Spatial pattern overlap was 0.74, with EVALVE showing a slightly broader highstress band along the free edge. Waveform alignment was within RMS 1.5 mmHg and 2.0% of peak flow; geometry matched to one specimen within 0.31 mm freeedge length and 0.28 mm commissure span.

EVALVE: Peak systolic opening. GCIs were 2.6% (coaptation near zero) and 3.1% (peak stress); observed orders near 2.0. Tolerance tightening changed peak stress by 0.7%. The bench peak stress at Qmax (1.24 MPa) was predicted as 1.38 MPa, a 11.3% overprediction. Dice overlap was 0.80. RMS errors were 1.9 mmHg and 2.6% of peak flow. Geometry matching as above.

EVALVE: Rapid closure. GCIs were 3.8% (coaptation fraction) and 7.9% (stress); observed orders near 1.5. Additional coupling iterations reduced residuals but outputs changed by 2.1% when tolerances were tightened, reflecting sensitivity. The bench closure stress peak (1.62 MPa) was predicted as 1.23 MPa, a 24% underprediction; timing lag was 1.9 ms relative to the pressure reversal onset (outside target). Dice overlap was 0.70. RMS pressure mismatch during closure was 2.3 mmHg; the peak overshoot was 26 mmHg in simulation vs. 30 mmHg measured. Geometry matching within 0.33 mm; only one specimen simulated for closure.

PULSE: Diastolic sealing. GCIs were 2.0% for coaptation fraction and 5.6% for peak stress; observed orders near 1.9. Tightening coupling thresholds and subiterations changed outputs by 0.5%. Bench coaptation fraction (0.64) was predicted as 0.61, a 4.7% difference. Bench principal stress (0.83 MPa) was predicted as 0.93 MPa, a 12.0% overprediction. Dice overlap was 0.80. Waveform RMS mismatches after tuning

the Windkessel were 1.6 mmHg (pressure) and 2.1% of peak flow; geometry matched one specimen within 0.29 mm freeedge length.

PULSE: Peak systolic opening. GCIs were 2.3% (coaptation fraction) and 3.4% (stress); observed orders near 2.0. Tightening coupling thresholds changed peak stress by 0.7%. Bench peak stress (1.24 MPa) was predicted as 1.39 MPa, a 12.1% difference. Dice overlap was 0.79. RMS errors were 2.0 mmHg and 2.8% of peak flow; geometry matching as above.

PULSE: Rapid closure. GCIs were 4.2% (coaptation fraction) and 8.2% (stress); observed orders around 1.5. Tightening coupling iterations from 20 to 40 reduced residuals but changed peak stress by 2.2%, indicating residual sensitivity to interface stabilization. The bench closure stress peak (1.62 MPa) was predicted as 1.17 MPa, a 28% underprediction; timing lag was 2.1 ms. Dice overlap was 0.68. RMS pressure mismatch during closure was 2.4 mmHg; the simulated overshoot was 25 mmHg vs. 30 mmHg measured. Geometry matching within 0.32 mm; only one specimen simulated for closure.

IRI results across models indicated that 86–94% of parameter influence weight was sourced from as-tested measurements. For ALEAF, 91% of the influence weight came from bench waveforms, as-measured thickness, and biaxial material parameters; the remaining 9% was attributable to the dispersion parameter κ and Rayleigh damping coefficients taken from literature or tuned against ancillary tests. For EVALVE, 89% of the influence weight was from measurements; slightly larger reliance on literature defaults included numerical parameters of the repulsive contact potential. For PULSE, 88% of influence came from measurements; the Windkessel parameters were tuned to bench waveforms, with residual uncertainty contributing 4% of output variance.

Alignment to bench drivings and mounting was within target for diastole and systole across models, with RMS errors below 2 mmHg and 2–3% of peak flow and geometric alignment within 0.3 mm and 0.5–0.7 degrees. Closure alignment was slightly worse for EVALVE and PULSE due to limits of their interface treatments during rapid transients, with RMS errors up to 2.4 mmHg and peak overshoot underpredicted by 4–5 mmHg.

Representativeness of as-tested specimens was strongest in ALEAF, where mapping of thickness and freeedge length was most complete, and at least one matched specimen was used for diastole and systole. In EVALVE and PULSE, single specimen matching was completed for each mechanism, but multiple specimens were not simulated per mechanism.

The three model-specific tables that follow grade each factor–mechanism pair on the four-level scale with a concise basis statement.

15. Credibility summary table: ALEAF

The ALEAF model exhibited the strongest overall evidence, particularly under diastolic sealing. The table below lists all 18 entries for the six factors across the three mechanisms. Each level reflects the grading rules set a priori, and the basis line restates a numeric result presented in the text.

16. Credibility summary table: EVALVE

EVALVE performed well in smooth regimes but showed larger discrepancies during diastolic sealing and rapid closure, consistent with known limitations of regularized immersed interfaces for contact-dominated behavior. The 18 graded entries are listed below.

17. Credibility summary table: PULSE

PULSE's partitioned strategy provided solid alignment in diastole and systole after tuning the upstream circuit but struggled to capture the abrupt closure spike without additional stabilization. The following table lists the 18 entries.

18. Discussion

The credibility assessment demonstrates that state-of-the-art FSI models can reach decision-grade evidence for some mechanisms of aortic valve function but not yet for all. Among the three mechanisms, diastolic sealing is best captured. ALEAF and PULSE reproduced coaptation fractions within 5% absolute and principal stresses within 12–15%; EVALVE underpredicted principal stress by 14% but provided acceptable coaptation geometry. These outcomes support using ALEAF as primary evidence for diastolic coaptation and stress in the COU, with PULSE as supportive and EVALVE as corroborating for coaptation shape trends.

At peak systolic opening, all models predicted principal stresses within 12% of bench estimates, and the spatial pattern of high stress aligned (Dice 0.79–0.82). The flow regime is smooth, and stress levels are influenced by leaflet curvature, which the models capture from the as-modeled geometry. These results permit using all three models as contributing evidence for design ranking on systolic stress, with ALEAF preferred when tight solver controls are needed to demonstrate insensitivity to tolerances.

Rapid closure under waterhammer is the most challenging. All models underpredicted peak principal stress (18–28%), and timing lags of 0.5–2.1 ms were observed. The fast transient makes both the interface treatment and any rate-dependent material effects more important. While the models reproduced the pressure overshoot within 4–5 mmHg and captured the spatial localization of high stress (Dice 0.68–0.78), the magnitude underestimation suggests that current modeling choices—contact regularization stiffness, neglected viscous damping anisotropy, and the lack of a rate-dependent fiber model—limit fidelity. For design ranking, model predictions during closure should be used as supportive evidence only, unless additional validation narrows the gap.

Regarding resolution effects across the ensemble, changes in coaptation fraction between the two finest space–time resolutions were at most 1.9% and changes in peak principal stress were at most 4.7%, indicating that the numerical resolution is not the dominant source of uncertainty for the two outputs under the tested conditions. This conclusion holds for diastole and systole and is somewhat weaker for closure, where local

nonmonotonicity appears but global quantities still approach convergence at the reported mesh/timestep combinations.

With respect to mathematical idealizations, bench comparison shows that model equations and constitutive choices are adequate for diastole and systole but underrepresent sharp transients during closure. Across models, signed biases in stress peaks ranged from 1.8% to 6.4% in diastole and systole (averaged across models where applicable) and from 18% to 28% during closure. The pattern of high stress is nevertheless colocalized along commissures and free edges, supporting qualitative use of spatial maps for design iteration.

On the parameter side, the mapping from bench measurements to model inputs was traceable and within specified tolerances. Between 86% and 94% of the total parameter influence weight came from astested measurements, including pressure/flow waveforms, specimen geometry, and material parameters; the remainder came from literature defaults or tuning of damping and dispersion. Sensitivity analysis showed that these residual parameters contributed less than 5% to the variance of the outputs in diastole and systole and up to 9% during closure.

Solver controls proved tight enough to render algorithmic choices a minor contributor to output variability under diastole and systole. Tightening nonlinear residual and coupling thresholds by a decade changed the outputs of interest by less than 0.7% across ALEAF and PULSE in these regimes; EVALVE showed slightly larger sensitivity under closure (up to 2.1%) due to immersed-interface iteration limits. After the reported settings, further tightening had no effect to three significant digits in ALEAF and minimal impact in PULSE.

The in vitro environment was mirrored with high fidelity: pressure and flow waveforms aligned to within RMS 1.5–2.4 mmHg and 2.1–2.8% of peak flow depending on mechanism, fixture angles within 0.4–0.7 degrees, axial mounting within 0.2–0.3 mm, and temperature within 0.6 °C. These small mismatches are unlikely to explain the closure stress underpredictions, which points instead to model form and high-rate material behavior limitations.

The simulated geometries corresponded to the astested units, with commissure-to-commissure span and leaflet free-edge length within 0.3 mm on average; remaining interspecimen variation ($n = 6$) accounts for a 2–5% spread in the outputs across diastole and systole based on a first-order propagation. This reinforces the practice of using at least one astested specimen per mechanism and encourages bracketing with two specimens when time allows, particularly for closure.

Our findings carry implications for the COU. ALEAF meets acceptance targets and achieves Level 3–4 evidence for diastole and systole, supporting its use as primary evidence to rank design variants. EVALVE provides corroborating results, especially for systolic stress distribution, but its underestimation of diastolic stress suggests using it as supportive evidence. PULSE achieves substantial evidence in smooth regimes and offers a practical platform for rapid design iteration given its partitioned architecture and ROM coupling; for closure, however, either improvements in interface stabilization and rate-dependent material modeling or additional validation are needed before it

can contribute decisively.

An ancillary consideration relates to software quality assurance. All three codes are under version control with regression tests for core modules, but we did not undertake an exhaustive independent software audit; future extensions of this work could expand the breadth of software process evidence alongside the physicsbased credibility activities.

Limitations include focus on a single valve size and platform; rigid aortic root walls; Newtonian fluid; and reliance on inverse FE for converting DIC displacements to principal stress, which introduces dependence on the chosen constitutive law. Nevertheless, the alignment between DICderived stress patterns and FSI predictions across mechanisms supports the internal consistency of the approach.

Practically, the credibility results guided design iteration. Across five candidate freeedge contours (not reported here in detail), ALEAF and PULSE simulations for diastole and systole produced a consistent ranking in coaptation and peak stress, leading to downselection of two variants for prototype fabrication. Closure predictions were used qualitatively to avoid designs with highly localized stress spikes near commissures, pending additional closurespecific validation.

19. Conclusions

Following ASME V&V 40, we assessed the credibility of three FSI models for predicting coaptation area fraction and peak principal stress under diastolic sealing, peak systolic opening, and rapid closure. Using a unified bench testing program for a 23mm stented bioprosthetic valve, we quantified verification metrics (GCIs, tolerance sensitivity) and validation metrics (absolute differences in the two outputs of interest, spatial pattern overlap, timing alignment) and graded six credibility factors for each model $\times$ mechanism pair. Across 54 entries, the strongest evidence was obtained for ALEAF in diastole and systole, with Level 4 resolution controls, Level 3–4 solver tightness, Level 3 mathematical idealizations, and Level 3 IRI; closure remained at Level 2 for model form due to underestimation of stress spikes. EVALVE and PULSE delivered Level 3 evidence in smooth regimes and Level 2 during closure.

For the COU—using simulations as primary evidence to rank candidate leaflet designs within the same platform—these results support the following practice: use ALEAF predictions as primary evidence for diastolic coaptation and systolic stress decisions; use PULSE for rapid iteration and as supportive evidence for diastole and systole; use EVALVE to corroborate systolic stress patterns and coaptation trends. For rapid closure, treat all three as supportive unless further mechanismspecific validation is performed or model enhancements (e.g., ratedependent fiber behavior, sharper contact regularization, and improved interface stabilization) close the magnitude gap.

The study demonstrates an endtoend, transparent, and quantitative credibility pathway that can be adapted to other cardiovascular devices and modeling approaches. By stating acceptance targets a priori, executing resolution and solver studies, tracing parameter provenance under the IRI, aligning

to bench drivings and mounting, and grading specimen representativeness, the evidence meets the expectations for highrisk model use without overclaiming in challenging regimes. Future work will extend the validation set to multiple valve sizes, flexible aortic roots, and enhanced closure modeling and will integrate a broader software process review to complement the physicsbased evidence.

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Table 1: Credibility assessment summary for ALEAF across three mechanisms.

Credibility factor	Level	Basis
Discretization error - ALEAF - Diastolic sealing	4	GCI: 1.3% (coaptation), 3.8% (stress); observed order 2.2; three-level refinement.
Model form - ALEAF - Diastolic sealing	3	Coaptation fraction 0.66 vs. 0.64 bench (2.9% diff); stress 0.91 vs. 0.83 MPa (9.4%); Dice 0.86.
Input Realism Index (IRI) - ALEAF - Diastolic sealing	3	91% of parameter influence weight from as-tested inputs; remaining from κ , damping with sensitivity <4%.
Numerical solver error - ALEAF - Diastolic sealing	4	Tightening tolerances by 10 $\times$ changed coaptation by 0.2% and stress by 0.3%; interface residual <1e5 N/node.
Test conditions - ALEAF - Diastolic sealing	4	RMS mismatch 1.1 mmHg (pressure), flow ~0; mount within 0.2 mm axial and 0.4 $^\circ$ rotation.
Test samples - ALEAF - Diastolic sealing	3	As-modeled specimen matched within 0.25 mm (free edge) and 0.18 mm (commissure span); single unit simulated.
Discretization error - ALEAF - Peak systolic opening	3	GCI: 2.1% (coaptation), 4.1% (stress); observed order ~2.0; three resolutions.
Model form - ALEAF - Peak systolic opening	3	Peak stress 1.38 vs. 1.24 MPa (11% diff); Dice 0.82 at Qmax; timing aligned by construction.
Input Realism Index (IRI) - ALEAF - Peak systolic opening	3	90% of influence from measured waveforms/materials; remaining from dispersion and damping with minor impact.
Numerical solver error - ALEAF - Peak systolic opening	3	Tightening tolerances changed peak stress by 0.6%; residuals to 1e6; interface force balance within 1e5 N/node.
Test conditions - ALEAF - Peak systolic opening	3	RMS errors 1.7 mmHg and 2.4% of peak flow; geometry matched within 0.27 mm and 0.5 $^\circ$.
Test samples - ALEAF - Peak systolic opening	2	Single as-tested specimen geometry used; broader lot statistics not simulated under systole.
Discretization error - ALEAF - Rapid closure water-hammer	3	GCI: 2.6% (coaptation), 5.2% (stress); observed order 1.6; mild nonmonotonicity near contact onset.
Model form - ALEAF - Rapid closure water-hammer	2	Peak stress 1.33 vs. 1.62 MPa (18%); timing lag 0.5 ms; Dice 0.78.
Input Realism Index (IRI) - ALEAF - Rapid closure water-hammer	2	Rate-dependent damping and κ from literature; measured drivings; influence weight from measurements ~86%.
Numerical solver error - ALEAF - Rapid closure water-hammer	3	Tighter tolerances changed peak stress by 0.9%; Aitken-stabilized interface; residuals below targets.
Test conditions	3	Simulated overshoot 27 vs.

Table 2: Credibility assessment summary for EVALVE across three mechanisms.

Credibility factor	Level	Basis
Discretization error - EVALVE - Diastolic sealing	3	GCI: 2.4% (coaptation), 6.1% (stress); observed order ~1.9; AMR nonmonotonicity noted.
Model form - EVALVE - Diastolic sealing	2	Coaptation 0.60 vs. 0.64 (6.7% low); stress 0.71 vs. 0.83 MPa (14.2% low); Dice 0.74.
Input Realism Index (IRI) - EVALVE - Diastolic sealing	3	89% influence weight from measured inputs; contact potential parameters from literature with low sensitivity.
Numerical solver error - EVALVE - Diastolic sealing	3	Extra interface iterations changed QoIs by 0.8%; residuals met 1e4 normalized threshold.
Test conditions - EVALVE - Diastolic sealing	3	RMS errors 1.5 mmHg and 2.0% of peak flow; geometry matched within 0.31 mm and 0.6 $^\circ$.
Test samples - EVALVE - Diastolic sealing	2	Single specimen used; partial geometric matching; no multi-unit simulation.
Discretization error - EVALVE - Peak systolic opening	3	GCI: 2.6% (coaptation), 3.1% (stress); observed order ~2.0.
Model form - EVALVE - Peak systolic opening	3	Peak stress 1.38 vs. 1.24 MPa (11.3% high); Dice 0.80 at Qmax.
Input Realism Index (IRI) - EVALVE - Peak systolic opening	3	90% influence from measured waveforms/materials; minor defaults for interface regularization.
Numerical solver error - EVALVE - Peak systolic opening	3	Tolerance tightening shifted peak stress by 0.7%; interface residuals below 1e4 normalized.
Test conditions - EVALVE - Peak systolic opening	3	RMS errors 1.9 mmHg and 2.6% of peak flow; mount within 0.3 mm and 0.6 $^\circ$.
Test samples - EVALVE - Peak systolic opening	2	Specimen-specific mesh fit; single unit only.
Discretization error - EVALVE - Rapid closure water-hammer	2	GCI: 3.8% (coaptation), 7.9% (stress); observed order ~1.5; regridding near closure.
Model form - EVALVE - Rapid closure water-hammer	2	Peak stress 1.23 vs. 1.62 MPa (24%); timing lag 1.9 ms; Dice 0.70.
Input Realism Index (IRI) - EVALVE - Rapid closure water-hammer	2	Measured drivings but limited rate-dependent properties; influence from measurements ~87%.
Numerical solver error - EVALVE - Rapid closure water-hammer	2	Even with 30 iterations, tightening changed peak stress by 2.1%; sensitivity remains.
Test conditions - EVALVE - Rapid closure water-hammer	3	RMS 2.3 mmHg; overshoot 26 vs. 30 mmHg; geometry matched within 0.33 mm and 0.6 $^\circ$.
Test samples - EVALVE - Rapid closure water-hammer	2	Single closure specimen simulated; no statistical bracketing.

Table 3: Credibility assessment summary for PULSE across three mechanisms.

Credibility factor	Level	Basis
Discretization error - PULSE - Diastolic sealing	3	GCI: 2.0% (coaptation), 5.6% (stress); observed order ~1.9; three levels.
Model form - PULSE - Diastolic sealing	3	Coaptation 0.61 vs. 0.64 (4.7% diff); stress 0.93 vs. 0.83 MPa (12% high); Dice 0.80.
Input Realism Index (IRI) - PULSE - Diastolic sealing	3	88% influence from as-tested measurements; Windkessel fit residuals contribute ~4% of variance.
Numerical solver error - PULSE - Diastolic sealing	3	Tightening coupling from 1e3 to 5e4 changed QoIs by 0.5%; interface power balance within 0.5%.
Test conditions - PULSE - Diastolic sealing	3	RMS errors 1.6 mmHg and 2.1% of peak flow; geometry matched within 0.29 mm and 0.6°.
Test samples - PULSE - Diastolic sealing	2	Single specimen used; no multi-unit simulation.
Discretization error - PULSE - Peak systolic opening	3	GCI: 2.3% (coaptation), 3.4% (stress); observed order ~2.0.
Model form - PULSE - Peak systolic opening	3	Peak stress 1.39 vs. 1.24 MPa (12.1% high); Dice 0.79.
Input Realism Index (IRI) - PULSE - Peak systolic opening	3	89% influence from measured inputs; Windkessel parameters tuned to bench waveforms.
Numerical solver error - PULSE - Peak systolic opening	3	Tighter coupling changed peak stress by 0.7%; convergence within 5e4 in displacement/traction.
Test conditions - PULSE - Peak systolic opening	3	RMS errors 2.0 mmHg and 2.8% of peak flow; mounting within 0.3 mm and 0.7°.
Test samples - PULSE - Peak systolic opening	2	Specimen-specific geometry applied; single unit simulated.
Discretization error - PULSE - Rapid closure water-hammer	2	GCI: 4.2% (coaptation), 8.2% (stress); observed order ~1.5; added-mass effects.
Model form - PULSE - Rapid closure water-hammer	2	Peak stress 1.17 vs. 1.62 MPa (28%); timing lag 2.1 ms; Dice 0.68.
Input Realism Index (IRI) - PULSE - Rapid closure water-hammer	2	Measured drivings; limited high-rate properties; influence from measurements ~86%.
Numerical solver error - PULSE - Rapid closure water-hammer	2	Increasing sub-iterations 2040 changed peak stress by 2.2%; coupling sensitivity persists.
Test conditions - PULSE - Rapid closure water-hammer	3	RMS 2.4 mmHg; overshoot 25 vs. 30 mmHg; geometry within 0.32 mm and 0.7°.
Test samples - PULSE - Rapid closure water-hammer	2	One closure specimen simulated; no bracketing across n = 6.